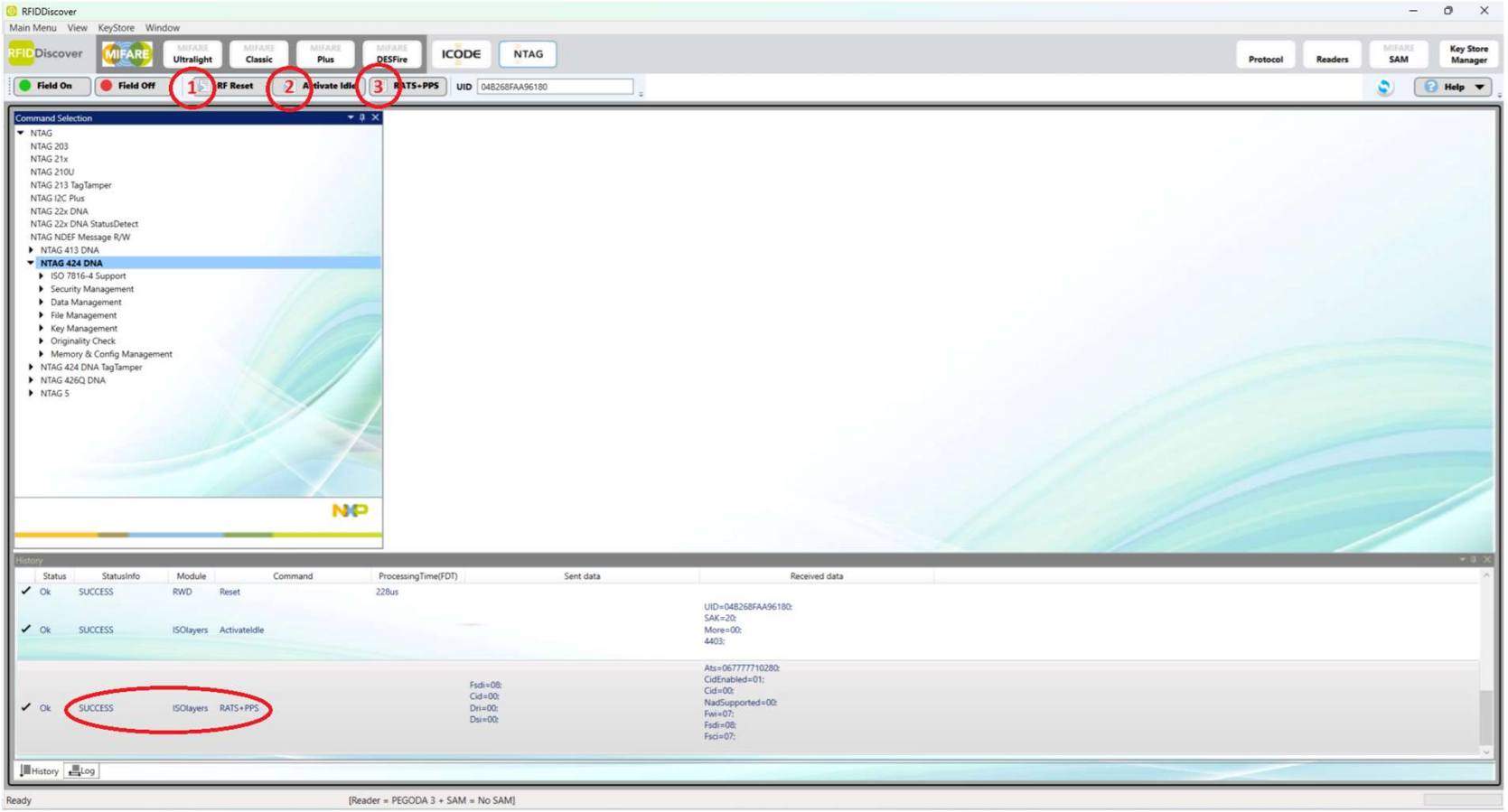
Personalization example in AN12196

Jimmy Chan
CAS TIC Connectivity Team
Support NFC, WIFI, BLE
April, 2025

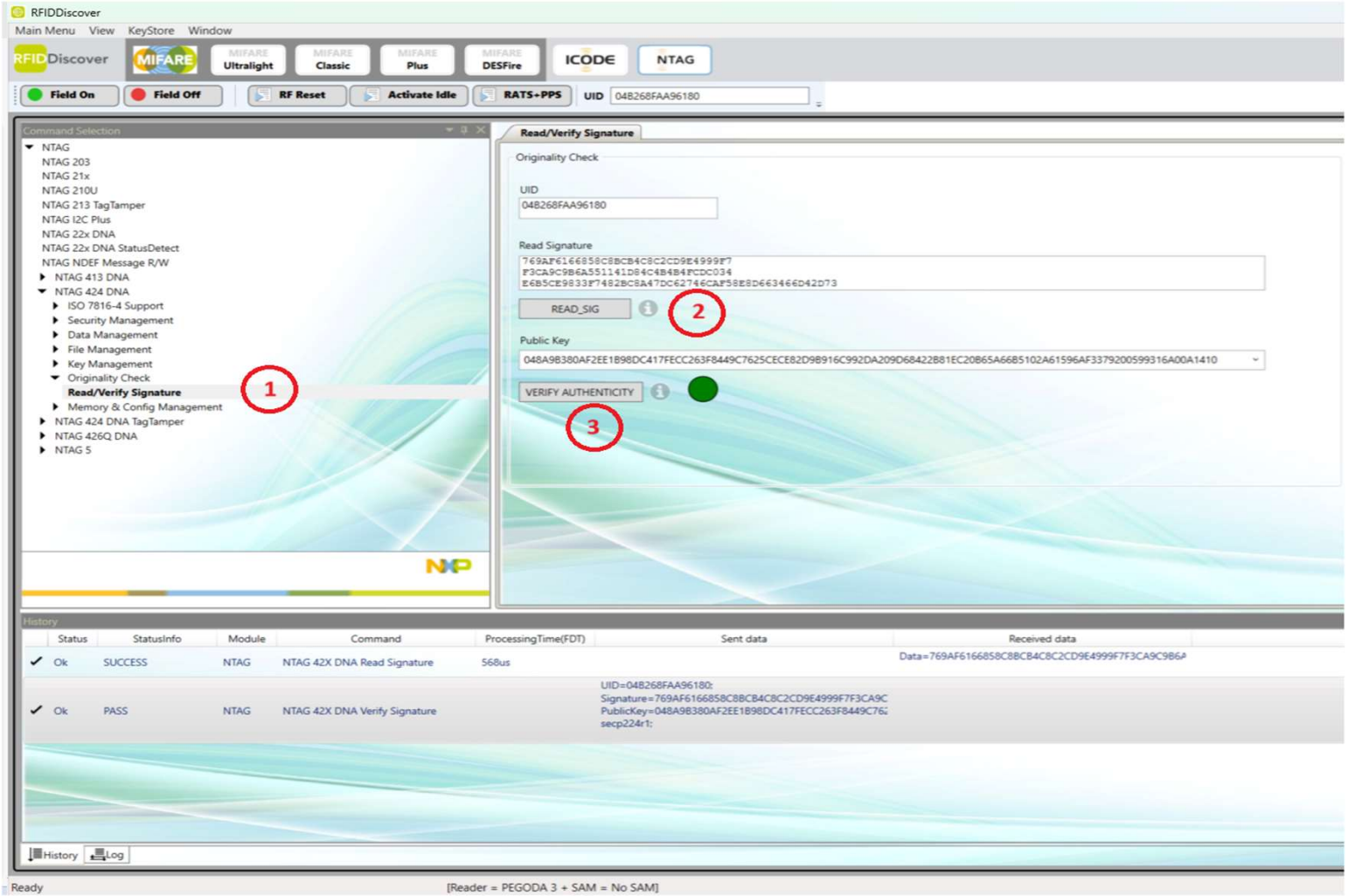
This document show the detail steps of following the Personalization example in AN12196. Using the Pegoda3 and RFIDDiscover.

ISO14443-4 PICC ACTIVATION

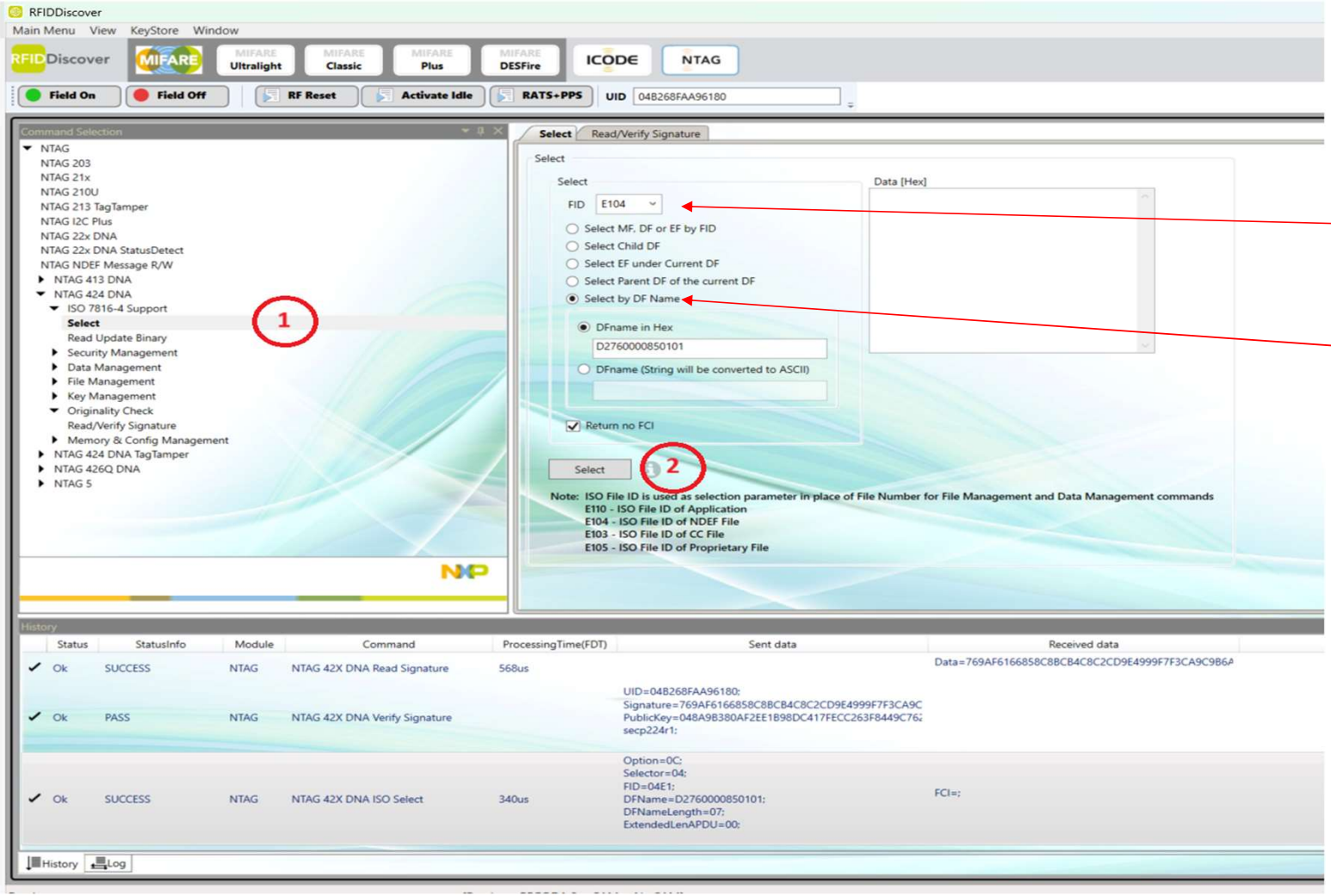
Press “RF Reset”, “Activate idle” and the “RATS+PPS”.



ORIGINALITY SIGNATURE VERIFICATION



ISO SELECT NDEF APPLICATION USING DF NAME



Select “E104”

Select by DF Name

GET FILE SETTINGS

RFIDDiscover

Main Menu View KeyStore Window

RFIDDiscover | MIFARE | MIFARE Ultralight | MIFARE Classic | MIFARE Plus | MIFARE DESFire | ICODE | NTAG

Field On | Field Off | RF Reset | Activate Idle | RATS+PPS | UID 04B268FAA96180

Command Selection

- NTAG
  - NTAG 203
  - NTAG 21x
  - NTAG 210U
  - NTAG 213 TagTamper
  - NTAG I2C Plus
  - NTAG 22x DNA
  - NTAG 22x DNA StatusDetect
  - NTAG NDEF Message R/W
  - NTAG 413 DNA
  - NTAG 424 DNA
    - ISO 7816-4 Support
      - Select
      - Read Update Binary
    - Security Management
      - Authentication
    - Data Management
    - File Management
      - **Get/Change File Settings**
      - Read NFC Counter
    - Key Management
    - Originality Check
      - Read/Verify Signature
    - Memory & Config Management
  - NTAG 424 DNA TagTamper
  - NTAG 426Q DNA
  - NTAG 5

1

Authentication | **Get/Change File Settings** | Select | Read/Verify Signature

000100

File Type: Standard Data File - NDEF File

SDM and Mirroring
UID Mirror
SDM Read Counter limit
SDM Read Counter
SDM ENC File Data
ASCII Encoding Mode

Read/Write Access Key: 00

Change Access Key: 00

Derivation Key for CMAC Calculation: --

SDM Ctr Ret Access Key: --

Mirroring Offset Bytes

VCUID Offset: 000000	MAC Input Offset: 000000	ENCPICC Data Offset: 1A0000
NFC Counter Offset: 000000	MAC Offset: 000000	Counter Limit Value: 000000
ENC Length: 000000	ENC Offset: 000000	

URI

Offset

Current Change :

Use Encryption

Get File Settings | 2 | Change File Settings

Note: ISO File ID is used as selection parameter in place of File Number for File Management and Data Management commands.
E110 - ISO File ID of Application.
E104 - ISO File ID of NDEF File.
E103 - ISO File ID of CC File
E105 - ISO File ID of Proprietary File

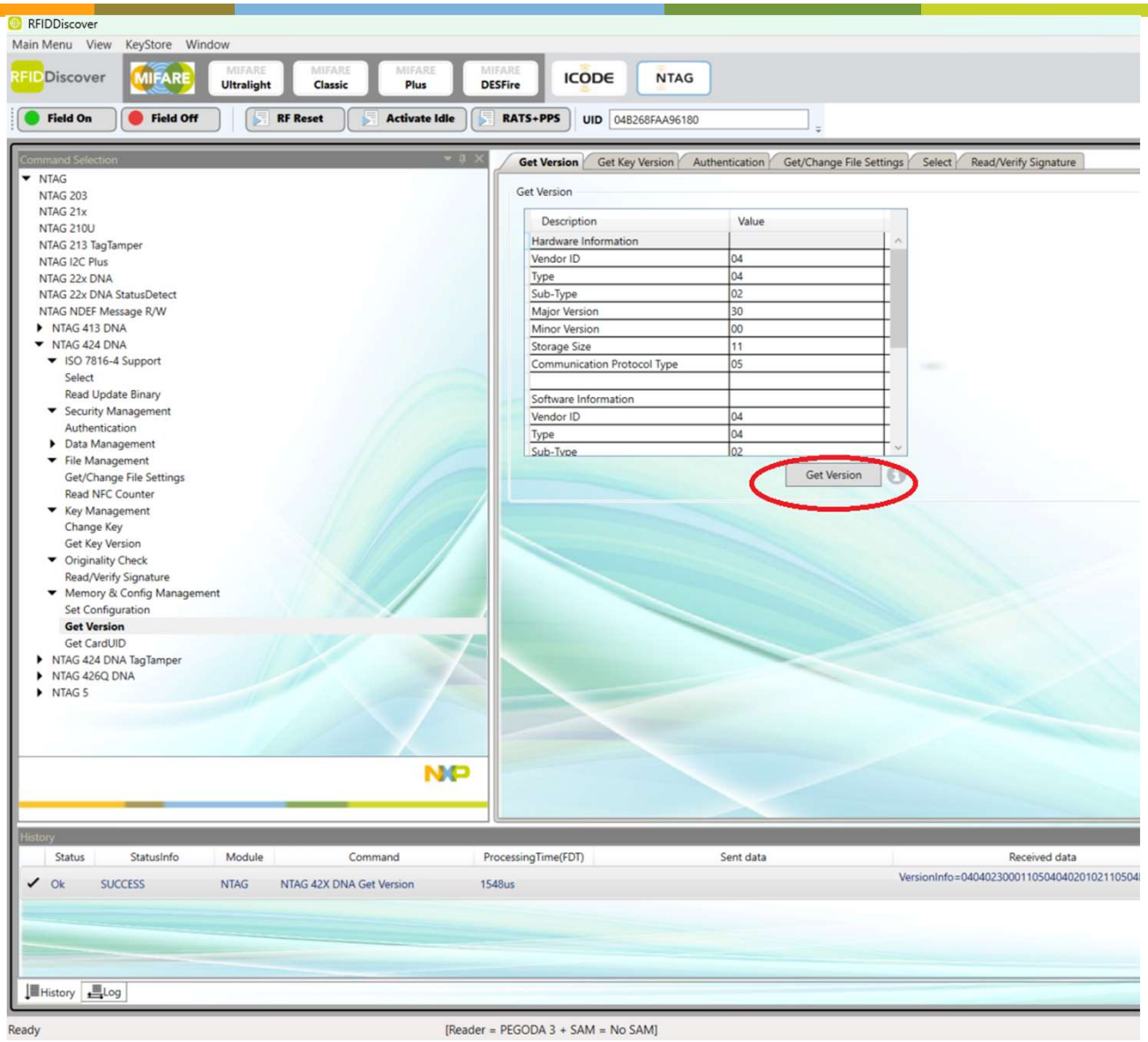
History

Status	StatusInfo	Module	Command	ProcessingTime(FDT)	Sent data	Received data
Ok	SUCCESS	NTAG	NTAG 42X DNA Change File Settings	5126us	Option=30; FileID=02; FileOption=00; AccessBytes=00E0;	
Ok	SUCCESS	NTAG	NTAG 42X DNA Get File Settings	991us	FileID=02;	FileSettings=000000E0000100;

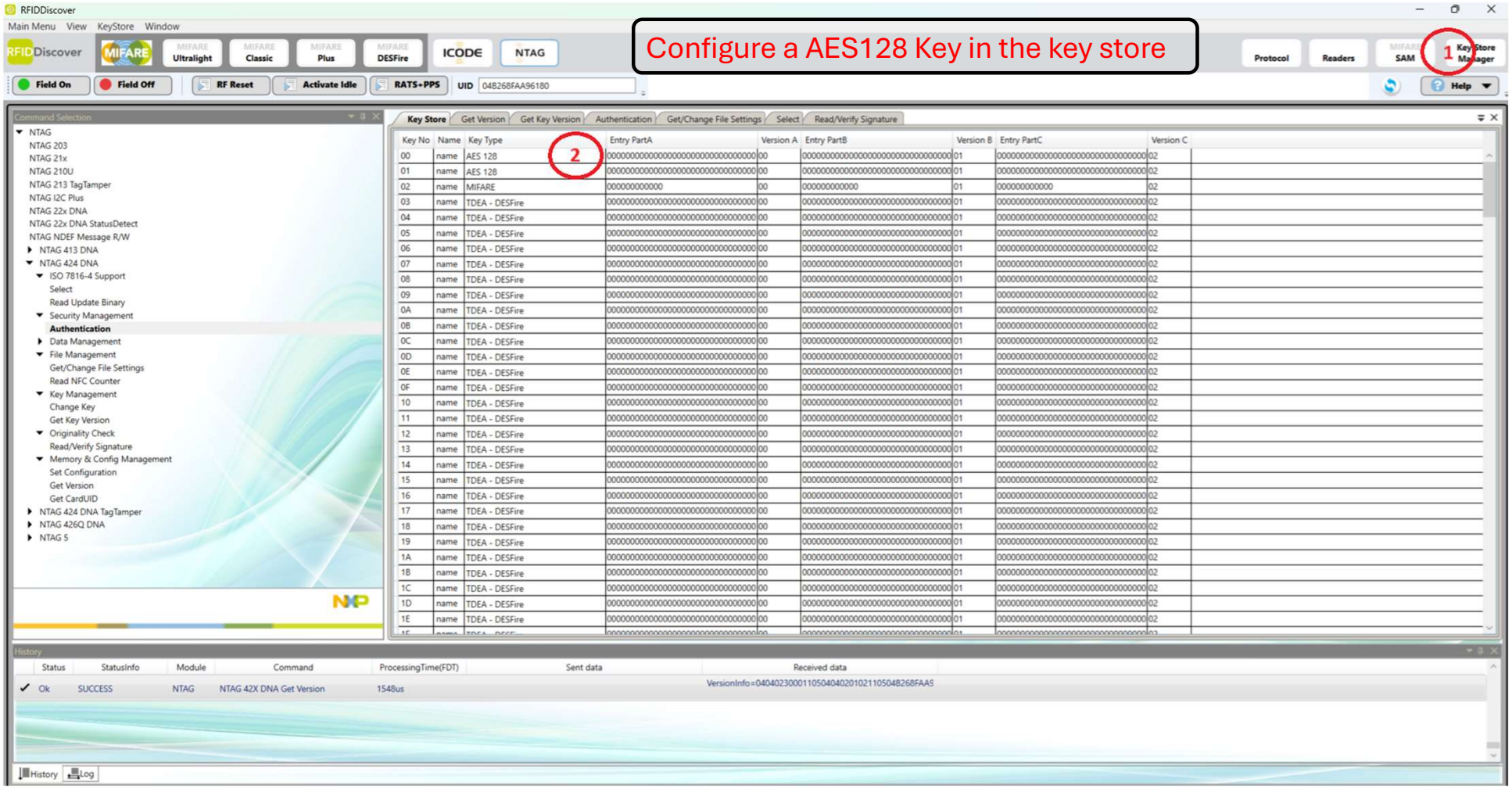
History | Log

Ready [Reader = PEGODA 3 + SAM = No SAM]

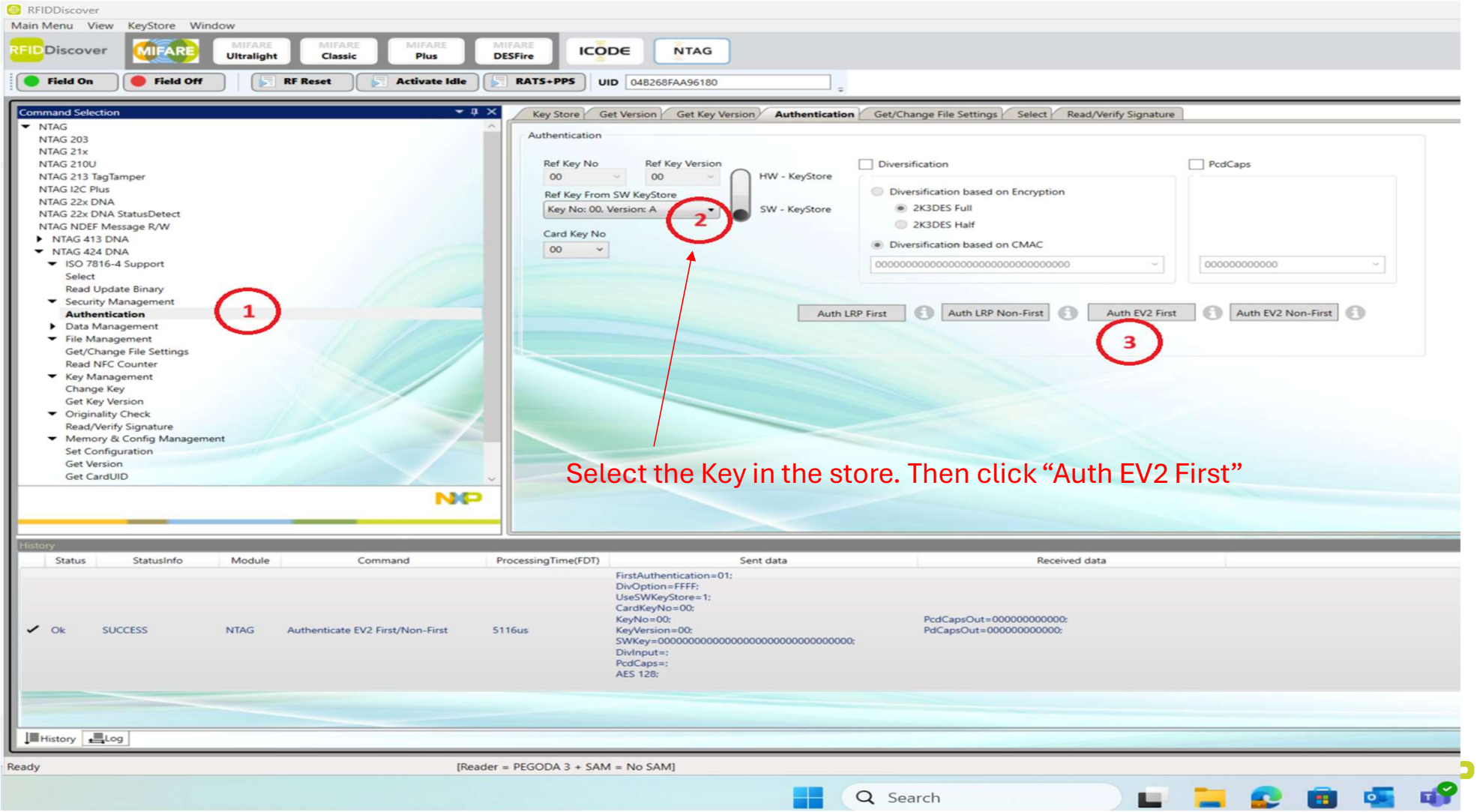
GETVERSION


RFIDDiscover
Main Menu View KeyStore Window
RFIDDiscover
MIFARE
MIFARE Ultralight
MIFARE Classic
MIFARE Plus
MIFARE DESFire
ICODE
NTAG
Field On
Field Off
RF Reset
Activate Idle
RATS+PPS
UID 04B268FAA96180
Command Selection
NTAG
NTAG 203
NTAG 21x
NTAG 210U
NTAG 213 TagTamper
NTAG I2C Plus
NTAG 22x DNA
NTAG 22x DNA StatusDetect
NTAG NDEF Message R/W
NTAG 413 DNA
NTAG 424 DNA
ISO 7816-4 Support
Select
Read Update Binary
Security Management
Authentication
Data Management
File Management
Get/Change File Settings
Read NFC Counter
Key Management
Change Key
Get Key Version
Originality Check
Read/Verify Signature
Memory & Config Management
Set Configuration
Get Version
Get CardUID
NTAG 424 DNA TagTamper
NTAG 426Q DNA
NTAG 5
Get Version
Get Key Version
Authentication
Get/Change File Settings
Select
Read/Verify Signature
Get Version
Description
Value
Hardware Information
Vendor ID 04
Type 04
Sub-Type 02
Major Version 30
Minor Version 00
Storage Size 11
Communication Protocol Type 05
Software Information
Vendor ID 04
Type 04
Sub-Type 02
Get Version
History
Status
StatusInfo
Module
Command
ProcessingTime(FDT)
Sent data
Received data
Ok
SUCCESS
NTAG
NTAG 42X DNA Get Version
1548us
History
Log
Ready
[Reader = PEGODA 3 + SAM = No SAM]

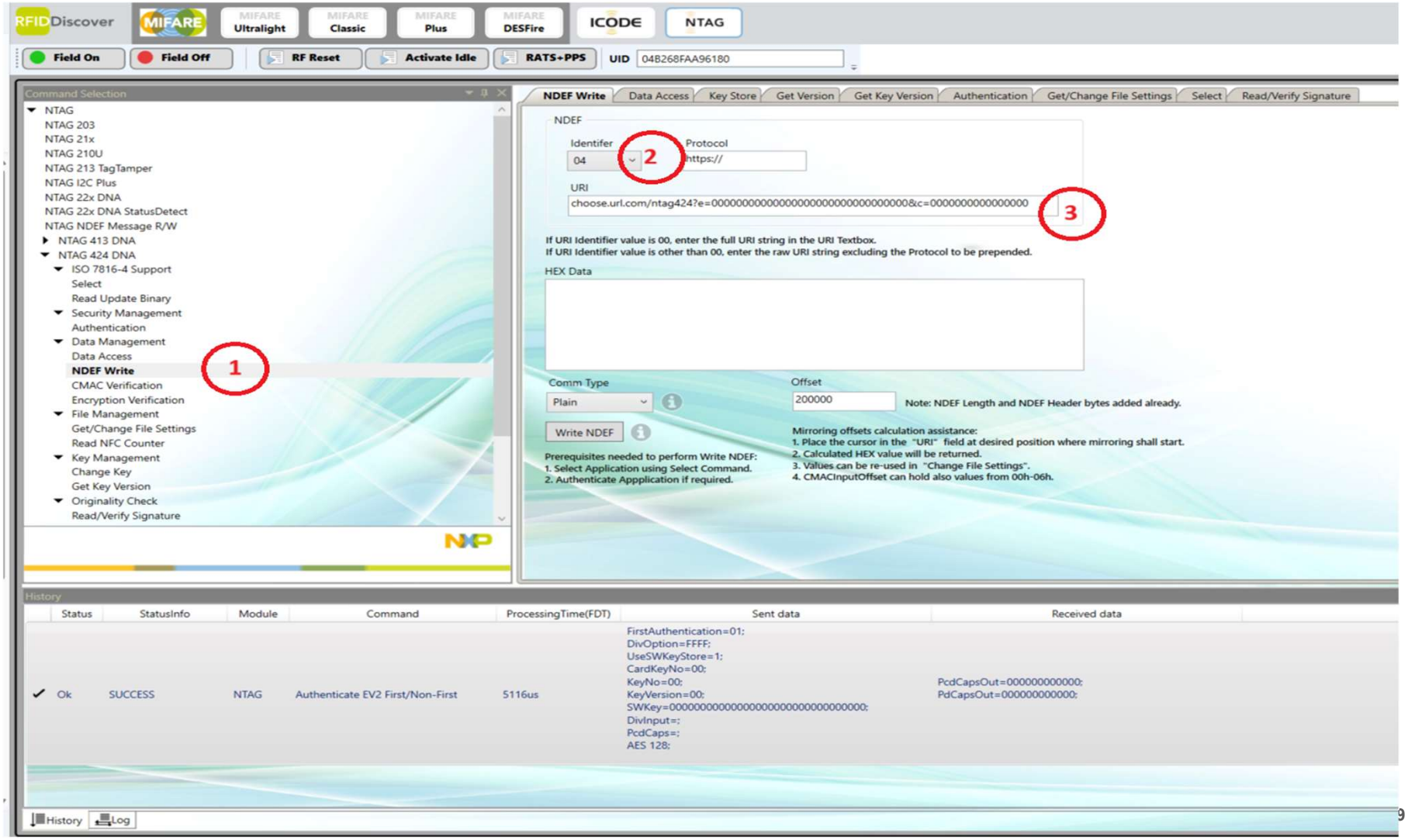
AUTHENTICATEFIRST WITH APPLICATIONKEY 0X00



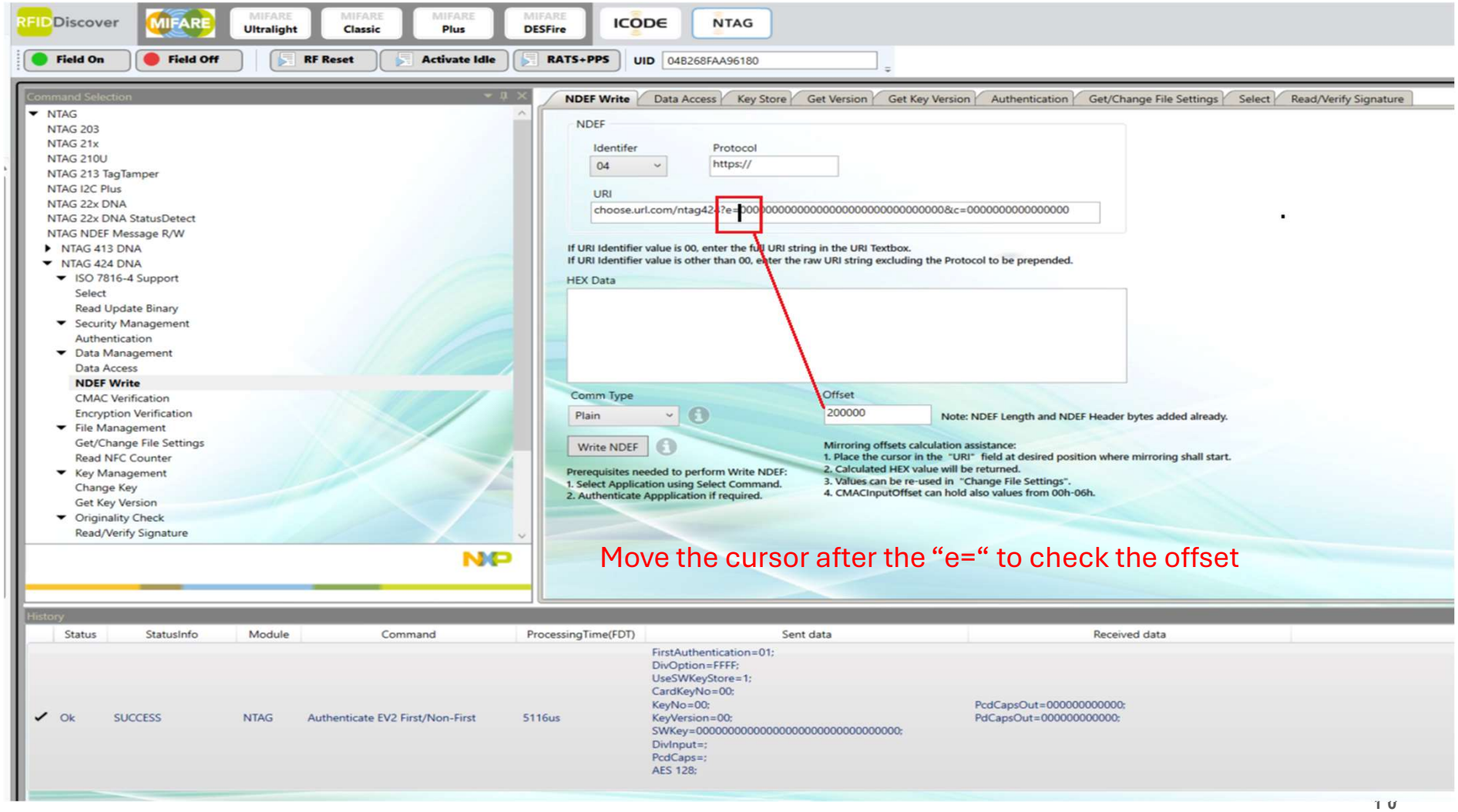
AUTHENTICATEFIRST WITH APPLICATIONKEY 0X00



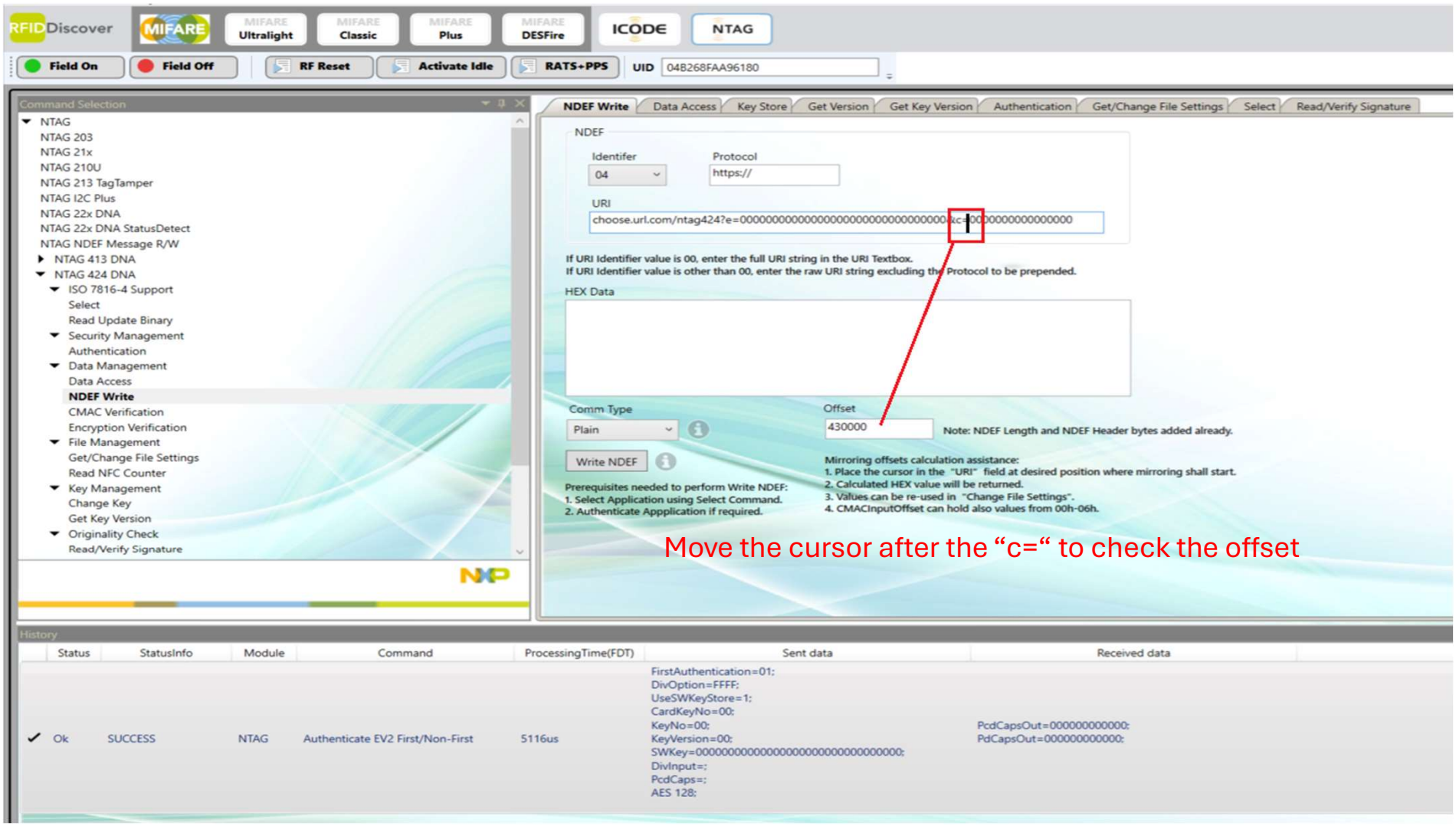
PREPARE NDEF DATA



PREPARE NDEF DATA



PREPARE NDEF DATA



WRITE NDEF

RFID Discover | MIFARE | MIFARE Ultralight | MIFARE Classic | MIFARE Plus | MIFARE DESFire | ICODE | NTAG

Field On | Field Off | RF Reset | Activate Idle | RATS+PPS | UID 04B268FAA96180

Command Selection

- NTAG
  - NTAG 203
  - NTAG 21x
  - NTAG 210U
  - NTAG 213 TagTamper
  - NTAG I2C Plus
  - NTAG 22x DNA
  - NTAG 22x DNA StatusDetect
  - NTAG NDEF Message R/W
  - NTAG 413 DNA
  - NTAG 424 DNA
    - ISO 7816-4 Support
      - Select
      - Read Update Binary
    - Security Management
      - Authentication
    - Data Management
      - Data Access
      - **NDEF Write**
      - CMAC Verification
      - Encryption Verification
    - File Management
      - Get/Change File Settings
      - Read NFC Counter
    - Key Management
      - Change Key
      - Get Key Version
    - Originality Check
      - Read/Verify Signature

NDEF Write | Data Access | Key Store | Get Version | Get Key Version | Authentication | Get/Change File Settings | Select | Read/Verify Signature

NDEF

Identifer: 04

Protocol: https://

URI: choose.url.com/ntag424?e=00000000000000000000000000000000&c=0000000000000000

If URI Identifier value is 00, enter the full URI string in the URI Textbox.
If URI Identifier value is other than 00, enter the raw URI string excluding the Protocol to be prepended.

HEX Data

0051D1014D550463686F6F73652E75726C2E636F6D2F6E7461673432343F653D30303030303030303030303030
303030303030303030303030303030303030303030303030303030303030303030303030303030303030303030

Comm Type: Plain

Offset: 430000

Note: NDEF Length and NDEF Header bytes added already.

Write NDEF

Prerequisites needed to perform Write NDEF:
1. Select Application using Select Command.
2. Authenticate Appplication if required.

Mirroring offsets calculation assistance:
1. Place the cursor in the "URI" field at desired position where mirroring shall start.
2. Calculated HEX value will be returned.
3. Values can be re-used in "Change File Settings".
4. CMACInputOffset can hold also values from 00h-06h.

History

Status	StatusInfo	Module	Command	ProcessingTime(FDT)	Sent data	Received data
✔ Ok	SUCCESS	NTAG	Write Data	4786us	CommMode=00; FileID=02; ISOOpt=01; Offset=000000; Length=530000; WriteDataBytes=0051D1014D550463686F6F73652E75726C	

CHANGE NDEF FILE SETTINGS

RFID Discover | MIFARE Ultralight | MIFARE Classic | MIFARE Plus | MIFARE DESFire | ICODE | NTAG

Field On | Field Off | RF Reset | Activate Idle | RATS+PPS | UID 04B268FAA96180

Command Selection

- NTAG
  - NTAG 203
  - NTAG 21x
  - NTAG 210U
  - NTAG 213 TagTamper
  - NTAG I2C Plus
  - NTAG 22x DNA
  - NTAG 22x DNA StatusDetect
  - NTAG NDEF Message R/W
  - NTAG 413 DNA
  - NTAG 424 DNA
    - ISO 7816-4 Support
      - Select
      - Read Update Binary
    - Security Management
      - Authentication
    - Data Management
      - Data Access
      - NDEF Write
      - CMAC Verification
      - Encryption Verification
    - File Management
      - **Get/Change File Settings** (1)
      - Read NFC Counter
    - Key Management
      - Change Key
      - Get Key Version
    - Originality Check
      - Read/Verify Signature
    - Memory & Config Management
      - Set Configuration
      - Get Version
      - Get CardUID
  - NTAG 424 DNA TagTamper

NDEF Write | Key Store | Get Version | Get Key Version | Authentication | **Get/Change File Settings** | Select | Read/Verify Signature

File Settings

File No: E104

Comm Type: Plain

File Size: 000100

File Type: Standard Data File - NDEF File

(2)
- [x] SDM and Mirroring
- [x] UID Mirror
- [ ] SDM Read Counter limit
- [x] SDM Read Counter
- [ ] SDM ENC File Data
- [x] ASCII Encoding Mode

Access Rights (3)

Read Access Key: 0E

Write Access Key: 00

Read/Write Access Key: 00

Change Access Key: 00

SDM Access Rights (4)

SDM Meta Read Access Key: 02

Derivation Key for CMAC Calculation: 02

SDM Ctr Ret Access Key: 01

Mirroring Offset Bytes

VCUID Offset: 000000

NFC Counter Offset: 000000

ENC Length: 000000

URI

MAC Input Offset (5): 430000

MAC Offset: 430000

ENC Offset: 000000

ENCPICC Data Offset (6): 200000

Counter Limit Value: 000000

Offset

Current Change :

The offset values are the position of “c=“ and “e=“ before.

History

Status	StatusInfo	Module	Command	ProcessingTime(FDT)	Sent data	Received data
✓ Ok	SUCCESS	NTAG	NTAG 42X DNA Change File Settings	6070us	Option=30; FileID=02; FileOption=40; AccessBytes=00E0; SDMOptions=C1; SDMAccessBytes=F122; SDMMACInputOffset=430000; SDMMACOffset=430000; EncryptedPICCDataOffset=200000;	

Before click “Change File Setting”, need to click “Auth EV2 First” again in Authentication

AUTHENTICATEEV2FIRST WITH KEY 0X03

RFIDDiscover | MIFARE Ultralight | MIFARE Classic | MIFARE Plus | MIFARE DESFire | ICODE | NTAG

Field On | Field Off | RF Reset | Activate Idle | RATS+PPS | UID 04B268FAA96180

Command Selection

- NTAG
  - NTAG 203
  - NTAG 21x
  - NTAG 210U
  - NTAG 213 TagTamper
  - NTAG I2C Plus
  - NTAG 22x DNA
  - NTAG 22x DNA StatusDetect
  - NTAG NDEF Message R/W
  - NTAG 413 DNA
  - NTAG 424 DNA
    - ISO 7816-4 Support
      - Select
      - Read Update Binary
    - Security Management
      - **Authentication**
    - Data Management
    - File Management
      - Get/Change File Settings
      - Read NFC Counter
    - Key Management
    - Originality Check
    - Memory & Config Management
  - NTAG 424 DNA TagTamper
  - NTAG 426Q DNA
  - NTAG 5

Read Update Binary | Select | **Authentication** | Get/Change File Settings

Authentication

Ref Key No: 00 | Ref Key Version: 00

Ref Key From SW KeyStore: Key No: 00, Version: A

HW - KeyStore / SW - KeyStore

Card Key No: 03 (1)

Diversification

Diversification based on Encryption
- 2K3DES Full
- 2K3DES Half

Diversification based on CMAC

00000000000000000000000000000000

PcdCaps

000000000000

Auth LRP First | Auth LRP Non-First | Auth EV2 First (2) | Auth EV2 Non-First

History

Status	StatusInfo	Module	Command	ProcessingTime(FDT)	Sent data	Received data
✓ Ok	SUCCESS	NTAG	Authenticate EV2 First/Non-First	5106us	FirstAuthentication=01; DivOption=FFFF; UseSWKeyStore=1; CardKeyNo=03; KeyNo=00; KeyVersion=00; SWKey=00000000000000000000000000000000; DivInput=; PcdCaps=; AES 128;	PcdCapsOut=000000000000; PdCapsOut=000000000000;

ISO SELECT PROPRIETARY FILE BY EF NAME

RFIDDiscover | MIFARE | MIFARE Ultralight | MIFARE Classic | MIFARE Plus | MIFARE DESFire | ICODE | NTAG

Field On | Field Off | RF Reset | Activate Idle | RATS+PPS | UID 04B268FAA96180

Command Selection

- NTAG
  - NTAG 203
  - NTAG 21x
  - NTAG 210U
  - NTAG 213 TagTamper
  - NTAG I2C Plus
  - NTAG 22x DNA
  - NTAG 22x DNA StatusDetect
  - NTAG NDEF Message R/W
  - NTAG 413 DNA
  - NTAG 424 DNA
    - ISO 7816-4 Support
      - **Select**
      - Read Update Binary
    - Security Management
      - Authentication
    - Data Management
    - File Management
      - Get/Change File Settings
      - Read NFC Counter
    - Key Management
    - Originality Check
    - Memory & Config Management
  - NTAG 424 DNA TagTamper
  - NTAG 426Q DNA
  - NTAG 5

Read Update Binary | **Select** | Authentication | Get/Change File Settings

Select

Select

FID E105 (1)

- (•) Select MF, DF or EF by FID
- ( ) Select Child DF
- ( ) Select EF under Current DF
- ( ) Select Parent DF of the current DF
- ( ) Select by DF Name
  - (•) DFname in Hex: D2760000850101
  - ( ) DFname (String will be converted to ASCII)

[x] Return no FCI

Select (2)

Data [Hex]

Note: ISO File ID is used as selection parameter in place of File Number for File Management and Data Management commands
E110 - ISO File ID of Application
E104 - ISO File ID of NDEF File
E103 - ISO File ID of CC File
E105 - ISO File ID of Proprietary File

History

Status	StatusInfo	Module	Command	ProcessingTime(FDT)	Sent data	Received data
✓ Ok	SUCCESS	NTAG	NTAG 42X DNA ISO Select	170us	Option=0C; Selector=00; FID=05E1; DFName=; DFNameLength=00; ExtendedLenAPDU=00;	FCI=;

ISO SELECT CC FILE BY EF NAME

RFIDDiscover

Main Menu View KeyStore Window

RFIDDiscover | MIFARE | MIFARE Ultralight | MIFARE Classic | MIFARE Plus | MIFARE DESFire | ICODE | NTAG

Field On | Field Off | RF Reset | Activate Idle | RATS+PPS | UID 04B268FAA96180

Command Selection

- NTAG
  - NTAG 203
  - NTAG 21x
  - NTAG 210U
  - NTAG 213 TagTamper
  - NTAG I2C Plus
  - NTAG 22x DNA
  - NTAG 22x DNA StatusDetect
  - NTAG NDEF Message R/W
  - NTAG 413 DNA
  - NTAG 424 DNA
    - ISO 7816-4 Support
      - Select
      - Read Update Binary
    - Security Management
      - Authentication
    - Data Management
      - Data Access
      - NDEF Write
      - CMAC Verification
      - Encryption Verification
    - File Management
      - Get/Change File Settings
      - Read NFC Counter
    - Key Management
      - Change Key
      - Get Key Version
    - Originality Check
      - Read/Verify Signature
    - Memory & Config Management
      - Set Configuration
      - Get Version

Encryption Verification | NDEF Write | Read NFC Counter | Get Key Version | Change Key | Read/Verify Signature | Data Access | Read U

Select

Select

FID E103

- Select MF, DF or EF by FID
- Select Child DF
- Select EF under Current DF
- Select Parent DF of the current DF
- Select by DF Name

1

- DFname in Hex: D2760000850101
- DFname (String will be converted to ASCII)

Return no FCI

Select 2

Data [Hex]

Note: ISO File ID is used as selection parameter in place of File Number for File Management and Data Management commands
E110 - ISO File ID of Application
E104 - ISO File ID of NDEF File
E103 - ISO File ID of CC File
E105 - ISO File ID of Proprietary File

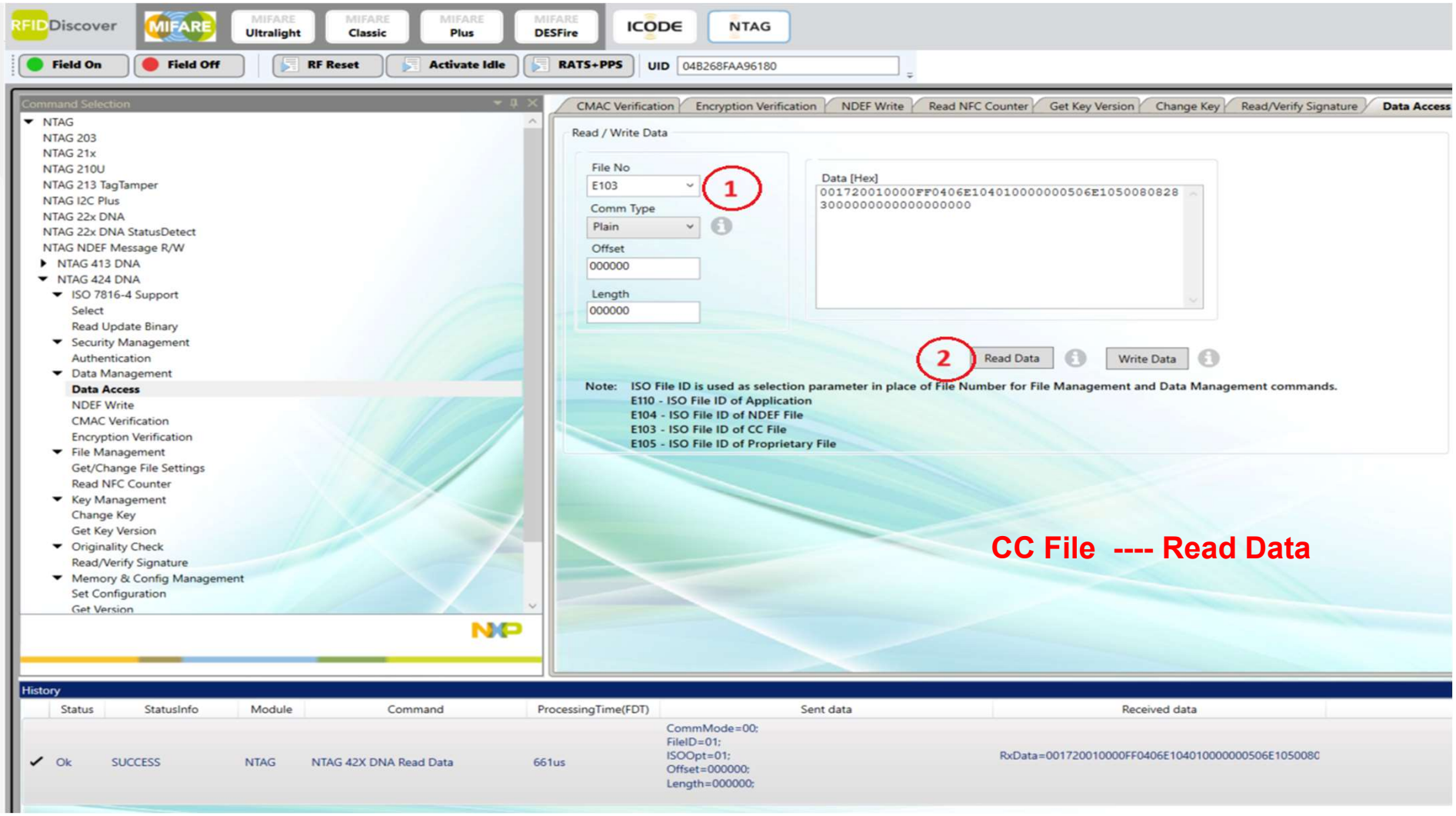
Log

```
phpall14443p4_Exchange--------ENTRY-------- wOption=C000  pTxBuffer=D2760000850101
phpall14443p4_Exchange--------LEAVE-------- [STATUS = SUCCESS]

Send to card: 00A4040C07D276000085010100

phpall14443p4_Exchange--------ENTRY-------- wOption=4000  pTxBuffer=00
phpall14443p4_Exchange--------LEAVE-------- ppRxBuffer=9000  [STATUS = SUCCESS]

Recv from card: 9000
phalMfNtag42XDna_IsoSelectFile--------LEAVE-------- [STATUS = SUCCESS]
```

You can find the APDU that send to card in the Log.

10	Cmd.Select C-APDU	>	00A4040C07D276000085010100

CONFIGURE CC FILE TO READ-ONLY



CONFIGURE CC FILE TO READ-ONLY

RFIDDiscover | MIFARE | MIFARE Ultralight | MIFARE Classic | MIFARE Plus | MIFARE DESFire | ICODE | NTAG

Field On | Field Off | RF Reset | Activate Idle | RATS+PPS | UID 04B268FAA96180

Command Selection

- NTAG
  - NTAG 203
  - NTAG 21x
  - NTAG 210U
  - NTAG 213 TagTamper
  - NTAG I2C Plus
  - NTAG 22x DNA
  - NTAG 22x DNA StatusDetect
  - NTAG NDEF Message R/W
  - NTAG 413 DNA
  - NTAG 424 DNA
    - ISO 7816-4 Support
      - Select
      - Read Update Binary
    - Security Management
      - Authentication
    - Data Management
      - **Data Access**
      - NDEF Write
      - CMAC Verification
      - Encryption Verification
    - File Management
      - Get/Change File Settings
      - Read NFC Counter
    - Key Management
      - Change Key
      - Get Key Version
    - Originality Check
      - Read/Verify Signature
    - Memory & Config Management
      - Set Configuration
      - Get Version

CMAC Verification | Encryption Verification | NDEF Write | Read NFC Counter | Get Key Version | Change Key | Read/Verify Signature | **Data Access**

Read / Write Data

File No: E103

Comm Type: Plain

Offset: 000000

Length: 000000

Data [Hex]

```
001720010000FF0406E104010000FF0506E1050080828
3000000000000000000
```

Read Data | Write Data

Note: ISO File ID is used as selection parameter in place of File Number for File Management and Data Management commands.
E110 - ISO File ID of Application
E104 - ISO File ID of NDEF File
E103 - ISO File ID of CC File
E105 - ISO File ID of Proprietary File

Change 00 to FF, then click "Write Data"

Log

```
phpall14443p4_Exchange--------ENTRY-------- wOption=C000   pTxBuffer=001720010000FF0406E104010000FF0506E1050080828300000000000000000
phpall14443p4_Exchange--------LEAVE-------- [STATUS = SUCCESS]

Send to card: 908D00002701000000200000001720010000FF0406E104010000FF0506E1050080828300000000000000000000

phpall14443p4_Exchange--------ENTRY-------- wOption=4000   pTxBuffer=00
phpall14443p4_Exchange--------LEAVE-------- ppRxBuffer=9100   [STATUS = SUCCESS]

Recv from card: 9100
phalMfNtag42XDna_WriteData--------LEAVE-------- [STATUS = SUCCESS]
```

SECURE CONNECTIONS
FOR A SMARTER WORLD

NXP and the NXP logo are trademarks of NXP B.V. All other product or service names are the property of their respective owners. © 2016 NXP B.V.